structures. Unlike mesh based geometric models, point clouds have no inherent surface or volume representation, and are relatively storage intensive. They also support taking precise measurements within the 3D model since the position of every point is stored with a high degree of precision, being based on the calculation of how long a beam of light from a laser takes to bounce off the physical object. Figure 2.13 shows a point cloud of the Morro Bay power plant in California, and how a measurement of the height of the stack can be taken in the model. You can explore this point cloud yourself by navigating to https://viewer.copc.io/?q=https://3d.dtia.site/pointcloud/ept.json.

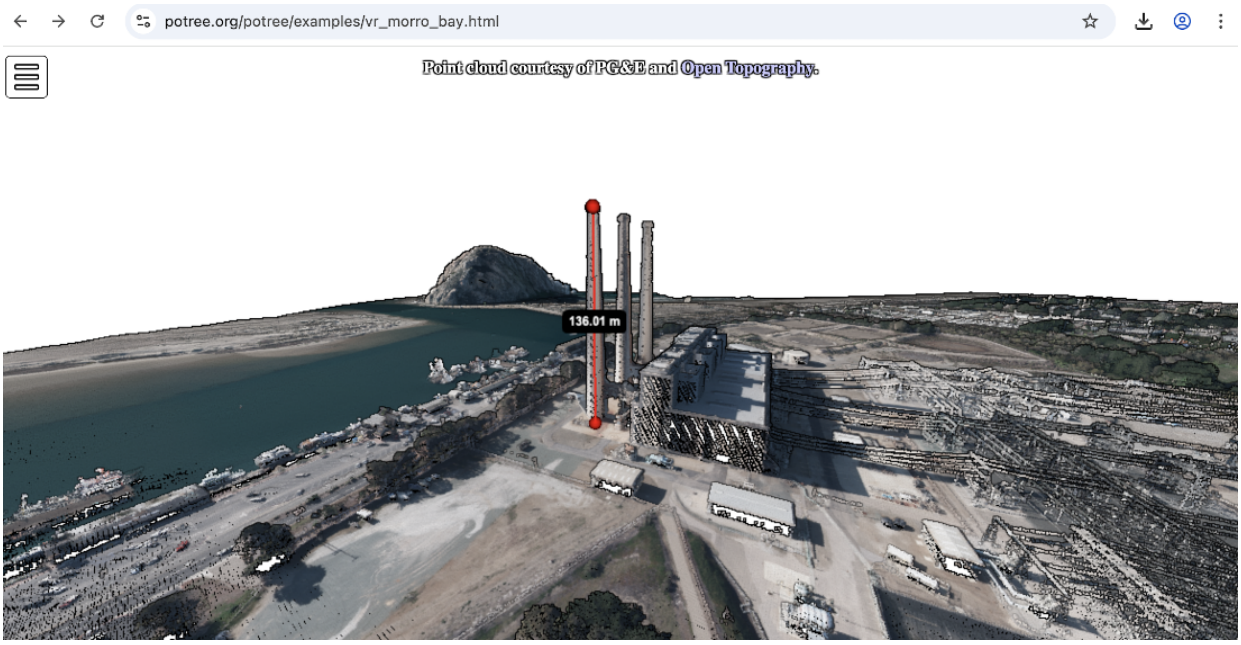


Figure 2.13 A point cloud visualization of the Morro Bay power station, showing how measurements can be taken within the point cloud. © PG&E Diablo Canyon Power Plant (DCPP): San Simeon and Cambria Faults, CA. Distributed by OpenTopography. https://doi.org/10.5069/G9CN71V5

OTHER TYPES OF 3D MODELS

Some domains require fundamentally different approaches to spatial representation. Medical imaging, materials science, and non-destructive testing have developed sophisticated techniques for capturing and modeling internal structures that remain invisible to conventional 3D scanning methods. Among these, computed tomography (CAT or CT scanning) and voxel-based modeling represent powerful paradigms that are also finding